

# Ashish Sukumar

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in ashish-sukumar

## Professional Summary

Robotics Engineering graduate student with hands-on experience in building, testing, and validating electromechanical systems. Skilled in integrating sensors, microcontrollers, and robotic subsystems, with strong experience in debugging, system-level troubleshooting, and prototype development. Comfortable working in lab environments with hardware, test setups, and real-world robotic systems. Strong collaborator with a detail-oriented approach to testing, documentation, and system reliability.

## Education

**Worcester Polytechnic Institute**

M.S. Robotics Engineering

GPA: 4.00 / 4.00

**SRM Institute of Science and Technology**

B.Tech Computer Science and Engineering

CGPA: 9.54 / 10

**Massachusetts, USA**

2025–Present

**Chennai, India**

2021–2025

## Technical Skills

**Languages:** Python, C++

**Hardware Platforms:** Arduino, ESP32, Raspberry Pi, NVIDIA Jetson

**Sensors:** IMU, LiDAR, Gas Sensors, IR, Ultrasonic, Camera Modules, Proximity Sensors

**Robotics:** ROS2, Webots, Gazebo, Motion Planning (OMPL)

**Systems:** Embedded Systems, Sensor Integration, Signal Processing, System Debugging

**Tools:** Linux, Git, MS Office

**Mechanical Exposure:** Basic CAD (Fusion 360), familiarity with mechanical assemblies and system integration

## Experience

**e-Yantra (IIT Bombay)**

Junior Project Technical Assistant

**Chennai, India**

Jun 2024–Feb 2025

- Assisted in building and testing robotic systems in simulation and hardware-integrated environments.
- Developed and validated **sensor-based perception modules**, ensuring reliability across different test scenarios.
- Supported debugging of system-level issues involving hardware-software interaction.
- Documented test procedures and system behavior for competition deployment.
- Collaborated with multidisciplinary teams to ensure task completion under strict timelines.

## Key Projects

**Fire Aware Smart-Bot – Autonomous Robotic System:** Designed and built an autonomous robot integrating multiple sensors (gas, IR, proximity) with microcontrollers (Arduino, ESP32) for real-time fire detection and navigation.

- Assembled and tested electromechanical components including motors and sensor modules.
- Performed system-level debugging to resolve communication and sensor noise issues.
- Conducted iterative testing to validate system performance under different environmental conditions.
- Implemented real-time control and monitoring through wireless communication.

**Fire Prediction System – Sensor + ML Integration:** Developed a fire prediction system using color sensors and machine learning models.

- Integrated hardware sensors with data acquisition pipelines for real-time inference.
- Tested and validated sensor readings across multiple scenarios to ensure robustness.
- Built supporting interface for monitoring predictions and system outputs.

**Robotics Simulation and Testing – Webots:** Developed and tested robotic task pipelines in simulation environments.

- Configured test environments to evaluate robot behaviors and task completion.
- Identified and debugged performance issues in navigation and perception modules.

- Assisted in validating system performance against expected task outcomes.

**Structure from Motion (SfM) – System Pipeline:** Built a multi-stage pipeline for 3D reconstruction from images.

- Focused on system integration, debugging, and validation of each pipeline stage.
- Strengthened ability to analyze complex system outputs and identify failure points.

## Publications

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Ashish, S., et al. (2025). *Fire Detection and Risk Prediction for Smart Safety*.

## Achievements

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- Red Hat Certified System Administrator (RHCSA)
- Oracle Cloud Infrastructure (OCI) Certified
- Excellence Award – SRM Institute of Science and Technology